

Document number	P0423R0
Date	2016-08-04
Project	Programming Language C++, Library Working Group
Reply-to	Jonathan Wakely <cxx@kayari.org>

Variable templates for Networking TS traits

The Networking TS defines type traits such as `is_executor` but does not provide corresponding variable templates such as `is_executor_v`. The direction set by C++17 is to provide these, so the TS should define them.

Proposed Wording

Add to 13.1 [async.synop] synopsis:

```
template<class T> struct is_executor;

template<class T>
    constexpr bool is_executor_v = is_executor<T>::value;

struct executor_arg_t { };
constexpr executor_arg_t executor_arg = executor_arg_t();

template<class T, class Executor> struct uses_executor;

template<class T, class Executor>
    constexpr bool uses_executor_v = uses_executor<T, Executor>::value;

template<class T, class Executor = system_executor>
    struct associated_executor;
```

Add to 16.1 [buffer.synop] synopsis:

```
// buffer type traits:

template<class T> is_mutable_buffer_sequence;
template<class T> is_const_buffer_sequence;
template<class T> is_dynamic_buffer;

template<class T>
    constexpr bool is_mutable_buffer_sequence_v = is_mutable_buffer_sequence<T>::value;
template<class T>
    constexpr bool is_const_buffer_sequence_v = is_const_buffer_sequence<T>::value;
template<class T>
    constexpr bool is_dynamic_buffer_v = is_dynamic_buffer<T>::value;

// buffer sequence access:
```